QP CODE: 25901165

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Value Addition Course(VAC)–MG4VACCND200-HEALTH AND WELLNESS

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S),Interest (I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A
Multiple ChoiceQuestions
Answer all questions
Each question carries **1** mark

1. Option B
2. Option C
3. Option B
4. Option D
5. Option C
6. Option A

Part B
Short Answer Type Questions.
Answer any **2** questions.
Each question carries **2** marks.

7. Explain the role of fiber

- Promotes bowel movement
- Prevents digestive disorders
- Controls blood sugar
- Lowers Cholestrol
- Provides Satiety

Answer any two points , each carries one mark

8. Marasmus usually affects infants and young children(below 1 year)who do not get enough breast milk or nutritious complementary foods.**(1 mark)**
It often occurs in infants under 1 year of age when breast feeding is stopped too early or milk is diluted or replaced with low-nutrient foods like watery rice gruel.**(1 mark)**

9.Explain the role of carbohydrates in our body

- Energy giving
- Protein sparing action
- Role in fat metabolism
- Storage of energy
- Structural and functional role
- Dietary fiber function
- Regulation of blood glucose

Answer any two points , each carries one mark

10. Explain Arrhythmias

- Type of cardiovascular disease
- Characteriised by irregular heart beats
- Dizziness, fainting or palpitations.

Answer any two points , each carries one mark

Part C
Short Essay Type Questions.
Answer any **4** questions.
Each question carries **5** marks.

11. Role of Vitamin A and Vitamin D

Role of Vitamin A

- Vitamin A is a fat-soluble vitamin essential for Vision,Growth and Development,Maintenance of Epithelial Tissue,Immunity and Reproduction **(1 mark)**
- Vitamin A deficiency leads to impaired vision and damage to the eyes and also affects growth,immunity and epithelial tissue health.**(1 mark)**
- Vitamin A deficiency results in eye problems,poor vision and increased risk of infections.**(1/2 mark)**

Role of Vitamin D

- Vitamin D is a fat soluble vitamin essential for the absorption of calcium and phosphorus from the intestine,which are vital for the growth and maintenance of healthy bones and teeth.**(1 mark)**
- Vitamin D deficiency leads to impaired bone mineralization and other health problems.**(1 mark)**
- VDD causes rickets in children and osteomalacia in adults.**(1/2 mark)**

12. Relationship between infections and Vitamin A Deficiency

- Vitamin A is called the Anti-infective vitamin.It is essential for maintaining a strong immune system and protecting the body from infections.**(1 mark)**
- When the body lacks Vitamin A,the immune system becomes weak,and the protective linings (epithelial tissues)of the eyes,lungs and intestines get damaged.**(1 mark)**
- In vitamin A deficient keratinised cells,the cilia of the epithelial cells are lost.**(1 mark)**
- This allows germs to enter the body more easily,increasing the risk of infections like measles,diarrhoea and respiratory diseases.**(1 mark)**
- VAD also reduces the number and activity of white blood cells, maker it harder for the body to fight infections.**(1 mark)**

13.Explain the health benefits of physical activity

I. Physical Health Benefits
- Improves cardiovascular health
- Supports healthy body weight
- Increases muscle strength and Endurance
- Enhances bone density
- Improves Respiratory function
- Regulates blood glucose levels
- Boosts Immunity
- Improve Digestive Health
- Reduces Risk of Chronic diseases

II. Mental and Emotional Benefits
- Enhances Mood
- Reduces stress and Anxiety
- Improves Sleep quality
- Boosts self esteem and confidence
- Enhances cognitive function
- Delays cognitive decline
- Promotes social well being

Answer any 10 points each carries ½

14.Explain the classification of Diabetes Mellitus

I. Type 1 Diabetes Mellitus :
Body doesnot produce insulin, caused due to autoimmune destruction of pancreatic beta cells. (2 marks)

III. Type 2 Diabetes Mellitus:
Body produces insulin but cells do not respond properly

IV. Gestational Diabetes Mellitus (2 marks)
Develops during pregnancy due to hormones that cause insulin resistance (1 mark)

15. Causes of Hypothyroidism

- **Iodine Deficiency-**Lack of iodine reduces thyroid hormone synthesis
- **Autoimmune thyroid disease(Hashimoto's thyroiditis)**-The body's immune system attacks the thyroid gland
- **Thyroid surgery or radiation therapy**-Reduces hormone production
- **Congenital hypothyroidism**-Present from birth due to defective or absent thyroid gland
- **Certain medications-**Such as Lithium and anti thyroid drugs
- **Pituitary or hypothalamic disorders**-Reduced TSH secretion leads to hypothyroidism
- **Inflammation or infection of the thyroid gland. (Any 5 points,1 mark for each point)**

16. Explain the functions and sources of protein.

- Growth and body building
- Repair and maintenance
- Formation of enzymes, hormones and antibodies
- Energy source
- Maintenance of fluid and acid base balance. (3 marks)

Sources: Animal sources includes milk, curd, eggs, meat, fish and poultry
Plant sources include pulses, legumes, soyabeans, nuts and oilseeds (2 marks)

Part D

Essay Questions
Answer any **2** questions
Each question carries **10** marks

17. Various consequences of under nutrition and the prevention methods

Consequences of under nutrition

- **Delayed physical and mental development in children-**slows down growth,resulting in stunted height,underweight and delayed puberty.Affects brain development,leading to poor cognitive function,learning difficulties and delayed speech and motor skills
- **Reduced immunity,increasing risk of infections-**Body more vulnerable to infections like diarrhoea,pneumonia and tuberculosis
- **Poor academic performance and productivity-**Children suffer from fatigue,lack of concentration and low energy levels.Adults-reduced physical strength and mental alertness,decreasing work capacity and productivity.
- **Increased maternal and infant mortality-** Under nourised mothers-higher risk of complications during pregnancy and child birth.Give birth to underweight or weak babies who are more likely to die in infancy or suffer from developmental issues.
- **Inter generational cycle of malnutrition-** Under nutrition passes from one generation to the next.Undernourished girls grow into undernourished women,who give birth to low birth weight babies,continuing the cycle of under nutrition in future generations.**(1 mark for each point)**

Prevention Methods:

- **Balanced diet rich in energy,protein,vitamins and minerals**
- **Exclusive breast feeding for the first 6 months and appropriate complementary feeding afterward**
- **Nutrition Education and awareness programs**
- **Food fortification(iodized salt,fortified flour,milk etc)**
- **Micronutrient supplementation(iron,folic acid,vitamin A)**
- **Improved sanitation and safe drinking water**
- **Poverty reduction and food security programs-**Govt.initiatives such as Public Distribution System(PDS),Mid day Meal Scheme and Integrated Child Development Services(ICDS) improve access to food and healthcare,reducing hunger and under nutrition **(Any 5 points,1 mark for each point)**

18. Explain the importance of fluid intake
    - Regulation of body temperature: sweating and evaporation (1 mark)
    - Facilitates digestion and Absorption: water is necessary for digestives juices that helps in the breakdown of food (1 marks)
    - Transportation of Nutrients and Oxygen: Water is the main component of blood and lymph (1 marks)
    - Excretion of Waste:Water helps kidneys to filter toxins and waste from the blood (1 marks)
    - Lubrication of Joints and Tissues: Water acts a lubricant and cushion for joints. (2 marks)
    - Cellular functions and metabolism: Water is required for metabolic reactions within cells (2 marks)
    - Protection of vital organs : Water provides a shock absorbing cushion for organs. (2 marks)

19. Explain the biological role of fats in our body.

    - Concentrated source of energy: Fats provide 9 kcal per gram
    - Supplies Essential fatty acids: Fats supplies linoleic acid and linolenic acids for normal growth and brain function.
    - Absorption of fat soluble vitamins: Dietary fats helps in the absorption of fat soluble vitamins
    - Insulation and Protection: Fats acts as insulating material, helping maintain body temperature.
    - Palatability and Satiety: Fats improve the taste and texture of foods.

(2x5=10 marks)

20. Explain the dietary management of diabetes

I Dietary Management:

A. Carbohydrate: Choose low GI foods, Increase fiber intake, Replace refined cereals with whole grains
B. Protein intake: moderate protein
C. Fat intake: Prefers healthy fats (MUFA, PUFA) **(3 marks)**

II Meal Planning Techniques

- Plate method- Half plate- non starchy vegetables, One fourth- Whole grains or millets, One fourth proteins, A small side fruit (low GI)

- Exchange lists: Food exchange lists
- Glycaemic load monitoring: Focus on both GI and portion size **(3 marks)**

III Foods to Include: vegetables, fruits, whole grains and millets, legumes and pulses, low fat milk/curd, Healthy nuts like almonds and walnuts

Foods to Avoid: Sweets, desserts, chocolates, white rice, fried items, chips, samosa, sweetened beverages, bakery foods containing trans fat **(4 marks)**